Which Side Is Wrong? Contradiction Flags in Retrieval-Augmented Generation Need Direction, Not Just Detection

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Abstract

Retrieval-augmented generation systems increasingly detect contradictions among retrieved passages and flag them to the generator. We ask what such a flag should say and what the detector must get right for it to help. In a pre-registered study on RAMDocs with two generators, we vary the detector (heuristic, NLI, LLM, random, oracle) and the format (neutral flag, directional flag naming the likely misinformation, or removal), and add synthetic detectors of controlled precision, recall and direction accuracy. A perfectly accurate neutral flag *lowers* accuracy on conflict questions by 6.7 points, because the model lists the misinformation alongside the correct answer; the same flag raises accuracy on genuine conflicts in WikiContradict. Naming the wrong side reverses the effect (+13.4 points, and +27.6 when the suspect passage is removed). Removal, however, is fragile: false flags delete correct passages, and with a real LLM detector its gains on conflict questions are cancelled on clean ones. Finally, pair-level detection F1 fails to predict downstream benefit, while a detector's document-level behaviour predicts it well. We recommend evaluating conflict detectors by direction accuracy and document-level effects on both conflict and conflict-free questions.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine: two sources report different findings, or two people share a name. Others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). A natural fix is to detect the contradiction and tell the generator about it. Recent systems pair pairwise contradiction detectors with prompts that flag conflicting passages (Wang et al., 2026), and prompting a model about conflicts can help (Cattan et al., 2025). These systems are judged mainly by detection F1 and end-to-end accuracy, which hides two questions. What should the flag say? And what must the detector get right for the flag to help?

We separate the two. We fix the retrieved passages and vary, independently, the *detector* (a heuristic, an NLI cross-encoder, an LLM, a random placebo and an oracle) and the *format* that passes its output to the generator: a neutral "these passages may conflict" flag, a *directional* flag that names the likely misinformation, or removal of the suspect passage. To see what detector quality buys, we also build synthetic detectors with a chosen precision, recall and direction accuracy. We ran pilots on 200 RAMDocs questions (Wang et al., 2025b), then pre-registered our hypotheses and tested them on 300 held-out questions with two generators, plus a contrast benchmark of genuine conflicts (Hou et al., 2024).

Our findings:

- **A correct neutral flag can hurt.** Flagging exactly the true contradictions lowers gpt-4o-mini's accuracy on RAMDocs conflict questions by 6.7 points [−10.8, −3.1]. The model reads "these conflict" as "both are valid answers" and repeats the misinformation. The same flag *raises* accuracy on WikiContradict, where both sides are valid, by 6.5 points (19.0 with gpt-4.1).
- **Direction is what helps.** Saying which passage is likely wrong gains 13.4 points on conflict questions, and removing that passage gains 27.6. A real gpt-4o-mini detector with directional output gains 12.5 (directional flag) and 18.7 (removal).
- **Removal is the high-variance choice.** The detector's false flags remove correct passages and cost 20.6 points on conflict-free questions, cancelling the gain overall (+0.3). A directional flag costs little on clean questions, so it is the safer interface.

- **Pair-level F1 does not predict benefit.** A detector's effect is predicted far better by what it does to *documents*: whether each question's misinformation is removed and whether a correct document is lost. The heuristic detector we started from does no better than random flags of the same volume.

Together these results suggest a reporting standard: evaluate a conflict detector by its document-level behaviour on both conflict and conflict-free questions, under the exact format it will feed the generator. We release the harness, pre-registration and every run.[1]

2 Related Work

Knowledge conflicts. Language models can give priority to memorised knowledge over contradicting context (Longpre et al., 2021). They are also readily persuaded by coherent external evidence that contradicts their memory (Xie et al., 2024), and they adopt wrong context over a correct prior to a degree that depends on how strong that prior is (Wu et al., 2024). Xu et al. (2024) distinguish context–memory, inter-context and intra-memory conflicts. We study inter-context conflicts: disagreements among retrieved passages, as in Chen et al. (2022), who find that models tend to rely on a single passage when the evidence conflicts.

Benchmarks with conflicting evidence. RAMDocs (Wang et al., 2025b) mixes genuine ambiguity (several valid answers), misinformation and noise across retrieved documents, and builds on AmbigDocs (Lee et al., 2024). CONFLICTS (Cattan et al., 2025) annotates search results with a taxonomy of conflict types, each with its own desired behaviour. WikiContradict (Hou et al., 2024) pairs Wikipedia passages that genuinely contradict each other. FaithEval (Ming et al., 2025), ConflictBank (Su et al., 2024) and ConfRAG (Yuan et al., 2026) cover related settings. We use RAMDocs because its document labels say which side of a contradiction is the misinformation, and WikiContradict as a contrast case where neither side is.

Telling the model about conflicts. The closest prior work tells the generator that a conflict exists. Cattan et al. (2025) find that prompting a model to reason about the conflict type helps, and that supplying the gold conflict type helps much more. On WikiContradict, models acknowledge both sides more often when told the passages may conflict (Hou et al., 2024). Ragability (Gross et al., 2026) shows the cost of such a hint: it improves conflict identification on contradictory contexts but hurts it on consistent ones. We extend these findings from conflict identification to answer accuracy. We separate the *content* of the hint (neutral, directional, removal) from its *accuracy* (precision, recall, direction), and we show that a neutral hint can hurt even when it is perfectly accurate.

Conflict detection and resolution in RAG. ConflictRAG (Wang et al., 2026) trains a pairwise detector on MiniLM embeddings with an LLM fallback, and reports pair-level detection F1 alongside answer gains. Its baselines include an NLI filter and a single-call chain-of-thought detector, both of which we reimplement in spirit. Gokul et al. (2025) find that LLMs are imperfect context validators whose accuracy varies with contradiction type. TCR (Ye et al., 2026) passes conflict signals to the generator through a learned soft prompt, and EvidentialRAG (Hossain et al., 2026) routes between answering and abstaining on fused evidential uncertainty. Credibility-aware generation (Pan et al., 2024) fine-tunes models to weigh per-passage credibility scores, the trained analogue of our directional flags. MADAM-RAG (Wang et al., 2025b) resolves conflicts implicitly, through debate among per-document agents. Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024) and Corrective RAG (Yan et al., 2024) assess passage reliability without an explicit contradiction detector. Our question is orthogonal to all of these: given a detector's output, which way of passing it to the generator helps, and what must the detector get right for that to happen?

Poisoned retrieval. A few injected passages can steer a RAG system's answer (Zou et al., 2025). Isolating passages and aggregating their answers gives certifiable robustness (Xiang et al., 2024). Our removal results connect to this line of work: removing the suspected passage is the strongest intervention we test, and also the one most damaged by false flags, which an attacker could trigger.

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

3 Framework

A question q comes with retrieved passages $P = (p_0, \ldots, p_{n-1})$. A *detector* returns flagged pairs $F \subseteq \{(i, j) : i < j\}$. A *directional* detector also returns, for each flagged pair, a *suspect* set $S_{ij} \subseteq \{i, j\}$: the side it believes is misinformation. A *format* turns (P, F, S) into the generator's prompt. We vary detectors and formats independently. Every prompt uses one skeleton: passages, question, and an instruction to end with `Final answer(s): a | b | ...`, listing every valid answer when different entities share a name (Appendix A).

3.1 Formats

No flags (plain). Passages and question only.

Instruction. Plain plus a generic instruction to check the passages for contradictions, with no detector. We test two wordings. v1 ends "if a disagreement reflects genuinely different valid answers, report each one"; v2 asks the model to judge which passages are reliable and to leave out answers supported only by unreliable ones.

Flag. Each flagged passage gets the header `[POTENTIAL CONFLICT with passage(s): j]`, followed by a list `Passages i and j: possible contradiction`. This is the format of the original CAAG system, except that every detector's flags are rendered identically (no detector-specific type labels or confidence scores), so detectors differ only in *which* pairs they flag.

Directional flag. As Flag, but suspect passages are headed `[LIKELY MISINFORMATION; conflicts with passage(s): j]`, and the list says which passage likely contains misinformation.

Remove (suspect). The suspect passages of every flagged pair are deleted from the prompt. For a pair without a suspect, the second passage is deleted, which is what the original CAAG REMOVE strategy does when there are no retrieval scores.

3.2 Detectors

CAAG (Contradiction-Aware Attention Gating) is a heuristic detector from an earlier, unpublished RAG system, used here with its code unchanged. A pair is scored only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity. A pair is flagged if the score reaches $\theta_c = 0.35$. It runs locally and gives no direction.

NLI is `nli-deberta-v3-base` (He et al., 2023), a cross-encoder scored in both directions; a pair is flagged if the larger contradiction probability exceeds 0.5. It gives no direction.

LLM is a single gpt-4o-mini call (temperature 0) that lists contradicting pairs, telling them apart from ambiguity, complementary and temporal relations, and names the side that is more likely misinformation (or "both" or "unclear"), as JSON.

Random flags each pair independently at CAAG's per-pair flag rate on the confirmatory split (6.22%). It is a placebo that matches CAAG's flag volume.

Oracle flags exactly the gold contradiction pairs and names the gold misinformation side.

3.3 Synthetic detectors

To separate a detector's accuracy from its identity, we build detectors with a chosen pair-level precision p, recall r and direction accuracy d from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a single per-pair rate, set over the whole split so that expected precision is p. That rate is dataset-wide, so conflict-free questions receive false flags too. A true flag names the gold misinformation side with probability d, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d, or lowering p, only adds flags or corrections. This keeps comparisons across cells paired.

3.4 Placing real detectors on the grid

Can the grid predict a real detector's effect from a few summary numbers? We compare two ways of summarising a detector. *Pair-level*: its pair precision, recall and direction accuracy. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document

marked; for clean questions, the share that have a correct document marked. Here "marked" means removed (Remove) or headed as misinformation (Directional flag). The prediction for a subset is the inverse-distance-weighted mean effect of the four nearest grid cells in that space.

4 Experimental Setup

Data. **RAMDocs** (Wang et al., 2025b) has 500 questions. Each has 1–12 documents (5.5 on average) typed *correct* (supports a gold answer, possibly for a different same-name entity), *misinfo* (supports a wrong answer) or *noise*. We label a pair a *contradiction* if one side is misinformation and the two answers differ, *ambiguity* if both are correct with different answers, *consistent* if the answers match, and *unrelated* if either is noise. A question is a *conflict* question if it has at least one contradiction pair, and *clean* otherwise.

The released types are imperfect. 163 documents typed *correct* contain a wrong answer; some are injected misinformation ("Ronald Powell, a renowned chess player"), others mention it incidentally (a navigation menu listing "Soccer"). We therefore leave pairs that touch these documents out of pair-level metrics, and drop the 55 questions whose only sign of conflict is such a document from the clean subset. Treating every such document as misinformation instead is a robustness check.

WikiContradict (Hou et al., 2024) has 253 questions, each with two Wikipedia passages that genuinely contradict each other. The reference answer is *both* sides and no passage is misinformation, so it is the contrast case to RAMDocs. **CONFLICTS** (Cattan et al., 2025): we use the 237 questions that have a reference answer (outdated-information, misinformation and no-conflict items), which can be scored automatically.

Pre-registration and splits. We developed the method on a random 200-question pilot split of RAMDocs. Those pilots motivated directional flags, removal and document-level placement. Before running anything on the remaining 300 questions (139 conflict, 126 clean, 35 excluded), we committed a pre-registration that fixes the conditions, metrics, hypotheses and corrections. Every RAMDocs result in the main text is on this confirmatory split. Pilot results appear only in the appendix and are marked exploratory.

Generators. The main generator is `gpt-4o-mini` (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We replicate the key conditions with `gpt-4.1`, keeping the LLM detector fixed at gpt-4o-mini. MADAM-RAG (Wang et al., 2025b) is ported unchanged to the same client: three debate rounds, temperature 0, 128 tokens per call.

Metrics. On RAMDocs an answer is correct if it includes every gold answer and no wrong answer (Wang et al., 2025b). Our primary metric reads "includes" literally: after normalisation (lowercasing; removing punctuation and articles), each gold or wrong answer must appear as a whole-word span of the extracted final answer. A more lenient bidirectional variant, which also counts a predicted answer contained in a gold one, is our robustness metric. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split (published: 28.00, on an undisclosed subset) and 32.8% on all 500 questions; with the bidirectional metric it scores 38.4% on all 500. We therefore do not claim to reproduce the published number exactly, and we compare conditions only within our own harness. On WikiContradict an answer is correct if it contains both sides; on CONFLICTS, if it contains the reference answer.

Statistics. We average each question's score over the three seeds, then report paired differences from the plain format with cluster-bootstrap 95% CIs (10,000 resamples) and two-sided sign-flip permutation tests (10,000 permutations) (Efron and Tibshirani, 1993; Dror et al., 2018). Holm correction (Holm, 1979) is applied within each pre-registered hypothesis family. A hypothesis is *robustly supported* if it also holds under the robustness metric.

Cost. Token counts come from the API's usage reports. Prices were checked on 2026-10-27 (gpt-4o-mini: \$0.15, \$0.075 cached, and \$0.60 per million input, cached-input and output tokens; gpt-4.1: \$2.00, \$0.50 and \$8.00). All experiments in this paper, pilots included, cost \$27.42 in API fees.

5 Results

Table 1 gives every condition on the RAMDocs confirmatory split, Table 3 gives the outcome of every pre-registered test, and Figure 1 plots each

Condition	All	Conflict	Clean	Tokens
No flags (plain)	29.9	22.1	41.5	911
Prompt only				
Instruction (v1)	+1.6	−1.2	+2.9	948
Instruction (v2)	+3.1	+5.3	−1.1	951
Oracle detector				
Oracle, flag	**−3.1**	**−6.7**	0.0	980
Oracle, directional flag	**+6.2**	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+27.6**	0.0	825
Real detectors				
Random, flag	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.7	−0.8	942
CAAG, remove	−3.3	0.0	−7.9	809
NLI, flag	+0.2	−0.5	+1.1	1021
NLI, remove	−5.7	−2.2	−8.2	592
LLM, flag	−1.8	−4.1	+1.9	970
LLM, directional flag	+5.0	+12.5	−3.4	983
LLM, remove suspect	+0.3	**+18.7**	**−20.6**	718
Reference				
MADAM-RAG	+5.7	+3.6	+7.9	8196

Table 1: RAMDocs hold-out (300 questions: 139 conflict, 126 clean), gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant after Holm correction across the table (sign-flip test, $p < 0.05$). Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

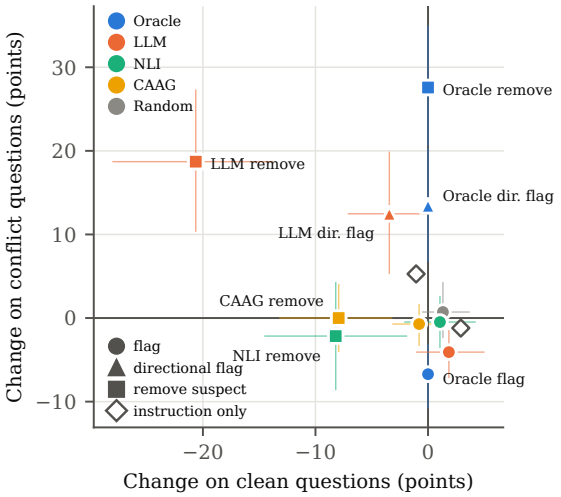


Figure 1: Change in one-way strict accuracy on RAMDocs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

condition's change on conflict questions against its change on clean ones.

5.1 A correct neutral flag hurts on misinformation and helps on genuine conflicts

Flagging exactly the gold contradictions, in the neutral format, *lowers* gpt-4o-mini's accuracy on conflict questions by 6.7 points [−10.8, −3.1], from 22.1% (H2). With gpt-4.1 the drop is 8.6 points (H9). Both hold under either metric. The errors are not refusals. With the flag, the share of conflict answers that include the misinformation answer rises from 47.7% to 61.9%, while gold recall barely moves (72.6% to 69.8%). The model lists the misinformation *alongside* the correct answer, treating "these passages conflict" as "both are valid". That reading is right when both sides are valid. On WikiContradict, where the reference answer is both sides, the same oracle flag *raises* the share of answers that give both sides by 6.5 points [+3.0, +10.0] with gpt-4o-mini, and by 19.0 points [+14.5, +23.8] with gpt-4.1 (H8; Table 9). A neutral flag passes on *that* a conflict exists, but not *which kind* it is. Whether it helps depends on whether listing both sides is the right behaviour.

5.2 Direction is what helps

Telling the model *which* side is likely wrong reverses the effect (H3): the share of answers that include the misinformation falls to 14.9% with the directional flag and 2.9% with removal. With oracle flags, the directional format gains 13.4 points on conflict questions [+6.7, +20.4], and removing the suspect passage gains 27.6 [+20.6, +35.0]. Both are robust to the metric and replicate with gpt-4.1 (+18.2 and +30.2; H9, Table 8).

5.3 What a detector must get right

Figure 2 replaces the oracle with synthetic detectors of known quality. Two thresholds appear. First, *direction accuracy*: at a direction accuracy of 0.5 (a coin flip on which side is wrong), no cell gains significantly in either format. Only perfect direction gives a significant gain on all questions (directional flag +6, removal +13 points at perfect precision). Second, *precision*, but only for removal: false flags make the model delete correct passages, which costs clean questions 15 points at precision 0.5 and 26 at 0.25. False directional flags cost clean questions at most two points. Across all grid cells, the harm on clean questions tracks the share of clean questions that lose a correct document (Spearman $\rho = -0.72$, $p < 0.001$), and 15 of 54 false-flag cells are significantly harmful

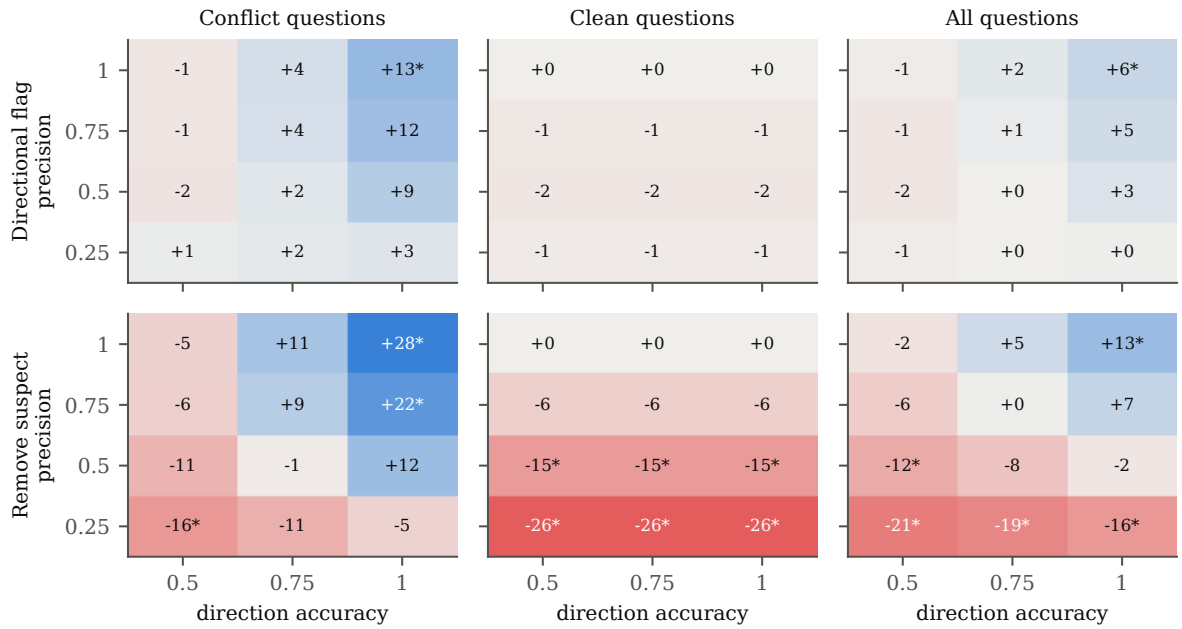


Figure 2: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction accuracy (columns), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that mode and subset. Clean questions only see false flags, so their rows do not vary with direction accuracy.

(H4). Removal is the more powerful and the more fragile interface: its best cell gains twice the directional flag's on all questions, but below precision 0.75 it is the only format that does net harm.

5.4 Real detectors

Table 2 describes the detectors. The pre-registered F1 ordering (H1) holds only in part. NLI beats CAAG (23.8 vs. 11.6), but the LLM detector's F1 (26.4) is not significantly higher than NLI's. Its direction accuracy is 88.3%, far above chance. NLI and the LLM detector flag most clean questions (61.9% and 76.2%); CAAG flags 34.1%.

The detectors' downstream effects differ far more than their F1 scores. In the neutral format, no real detector helps: CAAG, NLI and the LLM detector all stay within a few points of no flags. CAAG performs no differently from random flags at its own flag rate (H5; −0.4 vs. +1.0 points overall), so its pair selections carry no usable signal. Directional output changes the picture. The LLM detector gains 12.5 points on conflict questions with directional flags, and 18.7 with removal. Removal, however, costs 20.6 on clean questions, so its overall effect is +0.3. The directional flag costs 3.4 on clean questions and comes out +5.0 points ahead overall. With gpt-4.1 as the generator, the same gpt-4o-mini detector gives a significant overall gain with directional flags (+5.4 points, no loss on clean questions), while removal again nets out near zero (+1.0). For reference, MADAM-RAG gains 5.7 points overall at roughly nine times the prompt tokens. CAAG in its original typed and scored format behaves the same as in the uniform format (Appendix B).

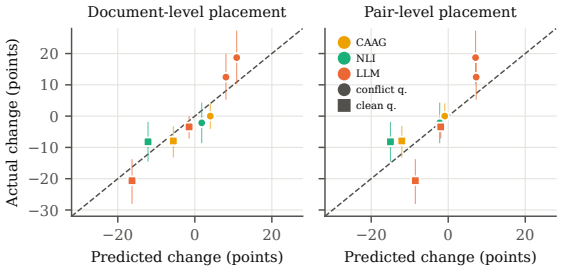


Figure 3: Effect of real detectors predicted from the synthetic grid against the measured effect (with 95% CIs), for conflict (●) and clean (■) questions. Left: detectors located by their document-level behaviour; right: by pair precision, recall and direction accuracy.

5.5 Document-level behaviour predicts a detector's effect; pair-level F1 does not

If the grid only described synthetic detectors, it would be of limited use. Figure 3 tests whether it predicts real ones (H6). Locating each real detec-

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 2: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

tor by its pair-level precision, recall and direction accuracy puts 5 of 8 effects inside their 95% CIs (mean absolute error 5.3 points). The LLM detector is the clearest failure: its pair recall is low, yet it removes the misinformation on 61.2% of conflict questions, because one correctly directed pair is enough to remove a misinformation document. Locating detectors by what they do to documents (the share of conflict questions whose misinformation is removed, and the share of clean questions that lose a correct document) puts 8 of 8 inside (error 4.1 points). We chose these axes after seeing the LLM detector on the pilot split. The confirmatory test covers three detectors and two formats.

5.6 Prompt-only baselines depend on wording

Neither instruction wording changes RAMDocs accuracy significantly after correction (H7). Their behaviour on WikiContradict explains why prompting alone is not a fix. The v1 instruction, which allows "genuinely different valid answers", raises both-sides answers there by 8.7 points; v2, which asks the model to drop unreliable passages, lowers them by 6.7. Each wording builds in one policy for handling conflicts, and that policy is right for one kind of conflict and wrong for the other. A directional detector can make that choice per pair.

CONFLICTS. On the 237 automatically scorable CONFLICTS questions (exploratory, one seed), no condition moves accuracy significantly; all are within ±5 points (Appendix B). Most of these items have no conflict (168), and 62 of the 69 conflict items involve outdated information.

6 Discussion

Why neutral flags backfire. A contradiction flag is a hint about *what to do* as much as a statement of fact. Our generation prompt, like most RAG prompts, asks for every valid answer when sources disagree for a legitimate reason. A neutral flag cannot separate that case from misinformation, so the model resolves the ambiguity toward inclusion. That is the right behaviour on WikiContradict and the wrong one on RAMDocs. The same tension appears in prompt-only baselines: each instruction wording builds in one policy and fails on the other kind of conflict. This sharpens the finding of Cattan et al. (2025) that knowing the conflict *type* helps. For misinformation, the useful part of the type is the direction.

Implications for building detectors. Detection F1, the usual figure of merit (Wang et al., 2026), measures the wrong unit. Over our three detectors, F1 ranged from 12 to 26 points, but the order of their downstream effects was set by two other properties. One is direction accuracy. The other is the detector's document-level footprint: whether every misinformation document in a question is caught, and whether a clean question loses a correct document. One correctly directed pair is enough to remove a misinformation document, while scattered false flags on correct–correct pairs (where most of our detectors' errors fall) delete valid answers. We recommend that conflict-aware RAG papers report (i) direction accuracy, (ii) document-level removal and loss rates, and (iii) effects on conflict-free questions, all under the exact format used downstream.

Which interface to use. Removal has the largest upside, roughly double the directional flag's at oracle quality, and the largest downside: it is the only format that does net harm below 0.75 precision, and with our LLM detector its gain on conflict questions is cancelled by losses on clean ones. A directional flag keeps most of the gain at a fraction of the risk, and it is the one interface that

gave a significant overall gain with a real detector (with the gpt-4.1 generator). Until detectors reach high precision, directional flags are the safer default. Removal should be gated on the detector's measured clean-question loss rate.

The heuristic detector. CAAG is the detector we set out to study. As a neutral flag it did no better than random flags of the same volume, and as a removal rule it only deleted correct passages (-7.9 points on clean questions, 0.0 on conflict ones). Its false flags fall mostly on pairs that should both be reported: 134 of its 221 false flags on the hold-out are same-name ambiguity pairs, which its entity-mismatch signal flags because they mention different dates or numbers. We read this as a warning about lexical contradiction cues in general (Gururangan et al., 2018), rather than as a problem with a particular threshold. Retuning illustrates why F1 misleads. Over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and $\theta_c = 0.10$. But it gets there by flagging 43% of all pairs and 87% of clean questions, which our grid predicts would be harmful under removal (Appendix B).

Limitations

Models. We test two closed OpenAI generators. Open-weights models may respond differently to flags, particularly to the directional header, and we could not run one within this study. The LLM detector is a single prompt to one model; stronger or fine-tuned detectors would occupy different points on the map.

Data. The confirmatory evidence is one benchmark, RAMDocs, with 300 questions. Its misinformation is synthetic and its document labels are imperfect: we exclude 55 questions whose status depends on 163 documents we could not label without human annotation, and we check that conclusions survive treating those documents as misinformation. WikiContradict and CONFLICTS each test one side of our account (genuine conflicts and real search results), but neither has a misinformation side that we can label at the passage level. We score CONFLICTS only on items with a reference answer and use no LLM judge, so its opinion and complementary-information items are not evaluated.

Metric. RAMDocs ships no scorer. Our primary metric reconstructs the published definition, and with it MADAM-RAG reproduces the published score within about one point on our pilot split. It is still string matching: it misses paraphrases, and partial answers interact with "no wrong answer". We report a more lenient variant throughout and mark any conclusion that depends on the choice.

Synthetic detectors. Synthetic false flags are spread uniformly over non-contradicting pairs, and their suspect side is chosen at random. Real detectors concentrate their errors, and the grid predicts them only once it is summarised at the document level. We chose those axes after seeing one detector on the pilot split; on the confirmatory split they are tested on three detectors, which is still a small sample.

Scope. We study flags passed to the generator as text. Learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may follow different trade-offs.

Ethics Statement

This work uses public benchmarks and commercial model APIs, and involves no human subjects or personal data. Its main risk is dual use. Our results show that a false "likely misinformation" label, or the removal of a passage, can change a RAG system's answers. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on conflict-free questions alongside every gain, and we argue that systems should measure what a detector does at the document level before acting on its flags.

References

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A Prompts

All prompts below are copied verbatim from the code that produced the results.

Generation skeleton.

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation
```

```
Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
[]}} if there are none.
```

B Additional Results

Robustness to the metric and the labels. Table 4 repeats Table 1 with the bidirectional metric. Table 5 repeats the pre-registered tests with every suspect document treated as misinformation. Table 3 lists all pre-registered tests.

Original CAAG format. Rendering CAAG's flags in its original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), changes accuracy by -0.9 points overall (-1.2 on conflict questions, -1.6 on clean ones). This is no different from the uniform format.

Pilot study (exploratory). On the 200-question pilot split, scored with the bidirectional metric, the same pattern appeared before any confirmatory run. Oracle neutral flags changed conflict-question accuracy by -10.5 points, directional flags by $+11.8$, and removing the suspect by $+28.4$. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents. They are why we added directional formats, synthetic direction accuracy and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 166 pairs in all. A single story rarely contradicts itself, so most of these flags are likely false; the flagged pairs are released for annotation. This is the setting where a false flag can only hurt.

C Reproducibility

Runs. Every number in this paper is produced by one script from committed run outputs: per-call responses, token counts, detector outputs and configurations. The pre-registration was committed before any confirmatory run. Its deviations are listed below.

Deviations from the pre-registration. (1) For the placement test (H6), the detectors without direction were run under the label `remove_suspect` as well as `remove`. Both render identical prompts, so these were served entirely from the response cache. (2) Table 1 applies Holm correction across all its comparisons. This is stricter than the per-family correction used for the hypothesis tests. (3) The pre-registration worded H6 as the actual effect lying inside a 95% CI around the grid prediction. The prediction has no bootstrap CI of its own, so we test the converse: whether the prediction lies inside the actual effect's 95% CI. The error comparison with pair-level placement is as pre-registered.

Cost and compute. All API calls in this paper, pilots included, cost $27.42. CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311	not supported
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001	supported
H1	direction accuracy llm_dir > 0.5	+38.3 [+31.9, +43.0]	<0.001	supported
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001	supported
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001	supported
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001	supported
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001	supported
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178	n.s.
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000	n.s.
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280	n.s.
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059	n.s.
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000	n.s.
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001	supported
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001	supported
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001	supported
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001	supported
H4a	false-flag cells below plain on clean q.	15 of 54 cells	Holm	supported
H4b	harm grows with clean-q. exposure (Spearman)	$\rho = -0.72$	<0.001	supported
H4c	removal harms clean q. more than directional flag	12 of 27 matched cells	Holm	supported
H6	document-level placement inside actual CI	8/8 (pair: 5/8)	MAE 4.1 vs 5.3	supported
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001	significant
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001	significant
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001	supported
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250	n.s.
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077	n.s.
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001	significant

Table 3: All pre-registered tests on the confirmatory split (differences in points of one-way strict accuracy unless stated). Holm correction within each family. Directional predictions are *supported* or *not supported*; tests with no predicted direction (H5, H7, and H8 apart from the oracle flag) are *significant* or *n.s.* †: holds under the primary metric but not under the bidirectional robustness metric.

Condition	All	Conflict	Clean	Tokens
No flags (plain)	35.2	25.9	49.7	911
Prompt only				
Instruction (v1)	+1.3	−1.0	+2.1	948
Instruction (v2)	+2.2	+3.6	−1.1	951
Oracle detector				
Oracle, flag	**−3.6**	**−7.7**	0.0	980
Oracle, directional flag	**+7.0**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+33.3**	0.0	825
Real detectors				
Random, flag	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.9	−0.8	942
CAAG, remove	−4.0	−1.0	**−8.5**	809
NLI, flag	+0.2	−1.4	+1.6	1021
NLI, remove	−8.0	−3.1	**−13.2**	592
LLM, flag	−3.4	**−7.2**	+0.8	970
LLM, directional flag	+4.7	**+14.1**	−6.1	983
LLM, remove suspect	−2.1	**+18.9**	**−26.5**	718
Reference				
MADAM-RAG	+5.7	+3.6	+6.9	8196

Table 4: RAMDocs hold-out (300 questions: 139 conflict, 126 clean), gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant after Holm correction across the table (sign-flip test, $p < 0.05$). Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	false-flag cells harmful on clean q.; Spearman ρ	15/54; $\rho = -0.51$	<0.001		

Table 5: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions).

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 6: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Conflict	No conflict
No flags (plain)	66.7	49.3	73.8
Instruction (v2)	−0.4	+1.4	−1.2
CAAG, flag	+0.4	+1.4	0.0
NLI, flag	−1.7	−4.3	−0.6
LLM, flag	+1.7	+2.9	+1.2
LLM, directional flag	−0.4	−2.9	+0.6

Table 7: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, one seed; exploratory. Contains-match accuracy (%) and differences in points; no difference is significant (bold would mark raw $p < 0.05$). CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	All	Conflict	Clean
No flags (plain)	32.7	21.8	48.4
Instruction (v2)	**+5.8**	**+12.2**	−2.6
Oracle, flag	**−4.0**	**−8.6**	0.0
Oracle, directional flag	**+8.4**	**+18.2**	0.0
Oracle, remove suspect	**+14.0**	**+30.2**	0.0
LLM, directional flag	**+5.4**	**+8.4**	0.0
LLM, remove suspect	+1.0	**+20.6**	**−24.3**
NLI, remove	**−7.0**	+0.2	**−15.6**

Table 8: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 1.

Condition	Both answers
No flags (plain)	14.0
Instruction (v1)	**+8.7**
Instruction (v2)	**−6.7**
Oracle, flag	**+6.5**
CAAG, flag	+0.7
NLI, flag	+3.2
LLM, flag	**+6.2**

Table 9: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), gpt-4o-mini, 3 seeds. First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant after Holm within H8.